SPARS setup guide

From a computer with nothing installed to an activated SPARS environment. Allow 20 to 30 minutes, most of it downloading.

1 Install Python

You need **Python 3.10 or newer** (3.11 or 3.12 is best). Check whether you already have it. Open a terminal (Windows: PowerShell) and run:

```
python --version        # Windows and most Linux
python3 --version       # macOS and some Linux
```

If it shows 3.10 or higher, skip to step 2. Anything older, or an error, means installing it. Run the line for your system, then close and reopen the terminal:

System	Command
Windows	`winget install --id Python.Python.3.12 -e`
macOS	`brew install python@3.12`
Linux	Install Python 3 and its venv module with your distribution's package manager.

> **Several Pythons installed?** The installer builds the environment with the Python that starts it. On Windows run `py -3.12 install.py` to choose a version; elsewhere `python3.12 install.py`.

2 Get the code

```
git clone https://github.com/RakaSP/SPARS-handson.git
cd SPARS-handson
```

This creates a folder called **SPARS-handson** and moves you into it. Every command from here on runs inside that folder.

> **No Git?** Open the repository page in your browser, choose Code, then Download ZIP. Unzip it, open a terminal in the unzipped folder and continue with step 3.

3 Run the installer

```
python install.py          # Windows and most Linux
python3 install.py         # macOS and some Linux
```

This prepares the environment SPARS needs. It takes a few minutes. When it succeeds, the last lines say **Setup complete.**

4 Activate the environment

The environment must be activated in every new terminal before you use SPARS. Run the line for your system, from inside the repository folder:

System / shell	Command
Windows, PowerShell	`.\SPARS-venv\Scripts\Activate.ps1`
Windows, Command Prompt	`SPARS-venv\Scripts\activate.bat`
macOS and Linux	`source SPARS-venv/bin/activate`

When it worked, the prompt starts with `(SPARS-venv)`. To leave the environment later, type `deactivate`.

5 Start the dashboard

```
python -m spars.cli 1
```

A full-screen dashboard opens with a list of experiment definitions. Press **q** to quit. The **1** is the number of simulation cores. You will use more in the session. Use a terminal window at least 120 columns wide.